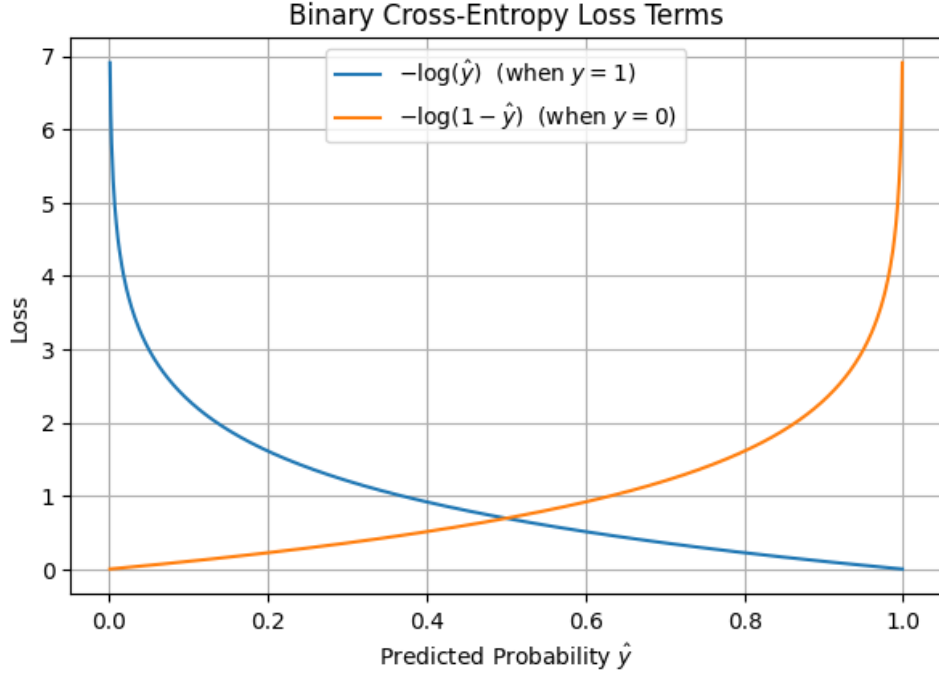




1 Gradient Derivation

1.1 Step 1: Differentiate BCE Loss

Starting with

$$\ell = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})], \quad (1)$$

differentiate with respect to $\hat{y}$:

$$\frac{\partial \ell}{\partial \hat{y}} = -\left(\frac{y}{\hat{y}} - \frac{1 - y}{1 - \hat{y}}\right). \quad (2)$$

1.2 Step 2: Differentiate Sigmoid

Since

$$\hat{y} = \sigma(z),$$

the derivative is

$$\frac{\partial \hat{y}}{\partial z} = \hat{y}(1 - \hat{y}). \quad (3)$$

1.3 Step 3: Apply Chain Rule

Using

$$\frac{\partial \ell}{\partial z} = \frac{\partial \ell}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z}, \quad (4)$$

we obtain

$$\frac{\partial \ell}{\partial z} = - \left(\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}} \right) \hat{y}(1-\hat{y}) \quad (5)$$

$$= \hat{y} - y. \quad (6)$$

Therefore,

$$\boxed{\frac{\partial \ell}{\partial z} = \hat{y} - y} \quad (7)$$

which is the prediction error.

1.4 Step 4: Differentiate Linear Model

Recall

$$z = \mathbf{w}^T \mathbf{x} + b. \quad (8)$$

Differentiating with respect to $\mathbf{w}$ gives

$$\frac{\partial z}{\partial \mathbf{w}} = \mathbf{x}. \quad (9)$$

Applying the chain rule once more,

$$\nabla_{\mathbf{w}} \ell = \frac{\partial \ell}{\partial z} \frac{\partial z}{\partial \mathbf{w}} \quad (10)$$

$$= (\hat{y} - y) \mathbf{x}. \quad (11)$$

Hence,

$$\boxed{\nabla_{\mathbf{w}} \ell = (\hat{y} - y) \mathbf{x}} \quad (12)$$

and

$$\boxed{\frac{\partial \ell}{\partial b} = \hat{y} - y.} \quad (13)$$

2 Gradient Descent Update Rule

The weight update equation becomes

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta(\hat{y} - y) \mathbf{x}. \quad (14)$$

Similarly,

$$b^{(t+1)} = b^{(t)} - \eta(\hat{y} - y). \quad (15)$$

The gradient consists of two components:

$$(\hat{y} - y) \mathbf{x}. \quad (16)$$

- $(\hat{y} - y)$ represents prediction error.
- $\mathbf{x}$ scales the update according to feature values.

Therefore,

- Large prediction error produces large updates.
- Small prediction error produces small updates.

3 Summary

The Logistic Regression model is

$$z = \mathbf{w}^T \mathbf{x} + b, \tag{17}$$

$$\hat{y} = \sigma(z), \tag{18}$$

$$\ell = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]. \tag{19}$$

The final gradients are

$$\boxed{\nabla_{\mathbf{w}} \ell = (\hat{y} - y) \mathbf{x}} \tag{20}$$

and

$$\boxed{\frac{\partial \ell}{\partial b} = \hat{y} - y.} \tag{21}$$

These equations form the foundation of Logistic Regression and modern Neural Network training using Gradient Descent and Backpropagation.